

Here are the complete lecture notes for "R-squared, Clearly Explained!!!".

You can copy everything below and paste it directly into a **new "Text" cell** in Google Colab. The math will render perfectly.

Lecture Notes: R-squared, Clearly Explained

I. Introduction to R-squared (R^2)

R-squared (or R^2) is a metric of correlation that is easy to compute and intuitive to interpret.

We are often already familiar with the standard metric for correlation, "plain old R."

- For **R**, values close to 1 or -1 indicate a *strong* relationship (e.g., weight and size).
- For **R**, values close to 0 indicate a *weak* relationship (lame).

Why do we need R^2 if we already have R?

The primary advantage of R^2 is its **ease of interpretation**.

- **Example:** Is a correlation of $R = 0.7$ "twice as good" as $R = 0.5$? It's not obvious.
- In contrast, R^2 values are directly proportional. An $R^2 = 0.7$ is literally 1.4 times as good as an $R^2 = 0.5$.
- R^2 is also very intuitive to calculate.

II. The Core Concept: Variation

To understand R^2 , we must first understand **variation**.

Step 1: The Baseline (Variation around the Mean)

Let's imagine a dataset where we plot mouse weight on the Y-axis.

1. First, we calculate the **mean** (average) of all the mouse weights. We can plot this as a flat, horizontal line across the graph.
2. Next, we measure how "spread out" the data is from this mean. This is the **Total Variation**.
3. We calculate this variation as the **Sum of the Squared Differences** between each data point (y_i) and the mean ($\bar{y}$).

The formula for this **Total Variation**, which we can call the **Sum of Squares Total** (SS_{total}), is:

$$SS_{total} = \sum_i (y_i - \bar{y})^2$$

- **Note:** We *square* the differences so that the points below the mean (which have a negative difference) don't cancel out the points above the mean (which have a positive difference).

Step 2: Adding a Predictive Model (The Regression Line)

Now, let's introduce a new variable, **Mouse Size**, on the X-axis.

- The data points are the same, they are just *reordered* by size.
- Because the Y-values haven't changed, the **Mean Line** and the **Total Variation** (SS_{total}) are *exactly the same as before*.

Key Question: Given that we know a mouse's *size*, is the mean line still the best way to predict its *weight*?

Answer: No. We can do much better by fitting a **regression line** (a sloped line) to the data. This new line allows us to make a much better prediction.

Step 3: Comparing the Models

How much *better* does our new regression line fit the data compared to the original mean line? **This is what R^2 measures.**

To quantify this, we need a new metric: the variation *around the new line*.

1. This is also a **Sum of Squared Differences**, but this time it's the difference between each actual data point (y_i) and the *predicted value on the regression line* ($\hat{y}_i$).
2. We can call this the **Sum of Squares Residual** ($SS_{residual}$). "Residual" means "left-over" variation that the line *doesn't* explain.

The formula for this **Residual Variation** is:

$$SS_{residual} = \sum_i (y_i - \hat{y}_i)^2$$

III. The Formula for R-squared (R^2)

R^2 is the **percentage of variation that is explained by the new line**.

We calculate it by finding out how much variation our new line *removed*.

- **Total Variation:** SS_{total} (Variation around the mean)
- **Unexplained Variation:** $SS_{residual}$ (Variation left over around the line)
- **Explained Variation:** $SS_{total} - SS_{residual}$

We express this as a percentage of the total:

$$R^2 = \frac{SS_{total} - SS_{residual}}{SS_{total}}$$

Or, written with the summation formulas:

$$R^2 = \frac{\sum (y_i - \bar{y})^2 - \sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$$

Properties of R^2 :

- It always ranges from **0 to 1** (or 0% to 100%).
- It's a **percentage**.
- R^2 can never be greater than 1, because the variation around the best-fit line ($SS_{residual}$) can never be greater than the variation around the mean line (SS_{total}).

IV. Worked Examples

Example 1: A Good Fit (Mouse Size vs. Weight)

Let's plug in some numbers for our mouse size vs. weight plot.

- Variation around the mean (SS_{total}) = **32**
- Variation around the line ($SS_{residual}$) = **6**

Calculation:

$$R^2 = \frac{32 - 6}{32}$$

$$R^2 = \frac{26}{32}$$

$$R^2 = 0.81 \text{ or } 81\%$$

Interpretation:

- This means there is **81% less variation** around the regression line than there was around the mean line.
- **Key Takeaway:** The "size-weight relationship" *accounts for 81%* of the total variation in weight.

Example 2: A Bad Fit (Sniffing a Rock vs. Weight)

Now, let's compare weight to a useless predictor: "time spent sniffing a rock".

- Variation around the mean (SS_{total}) = **32** (this never changes for the same Y-variable)
- Variation around the line ($SS_{residual}$) = **30**

Calculation:

$$R^2 = \frac{32 - 30}{32}$$

$$R^2 = \frac{2}{32}$$

$$R^2 = 0.06 \text{ or } 6\%$$

Interpretation:

- This means there is only **6% less variation** around the regression line.
- **Key Takeaway:** The "sniff-weight relationship" only *accounts for 6%* of the total variation. This is a very poor model.

V. How to Interpret R and R^2 in Practice**The R^2 Mindset**

- If someone says $R^2 = 0.9$ (or 90%):
 - **You think:** "Very good! The relationship between these two variables explains 90% of the variation in the data."
- If someone says $R^2 = 0.01$ (or 1%):
 - **You think:** "Dag! Who cares if this is 'statistically significant'? It only accounts for 1% of the variation. Something *else* must be explaining the other 99%."

The R vs. R^2 Mindset

R^2 is simply the **square of R**.

$$R^2 = (R)^2$$

This is why R^2 is much more intuitive.

- If someone says $R = 0.9$:
 - **You think:** " $0.9 \times 0.9 = 0.81$ ". This relationship explains **81%** of the variation.
- If someone says $R = 0.5$:
 - **You think:** " $0.5 \times 0.5 = 0.25$ ". This relationship explains **25%** of the variation.

This solves our original problem:

- "How much better is $R = 0.7$ than $R = 0.5$?"
- **Answer:**
 - $R_1 = 0.7 \rightarrow R_1^2 \approx 0.49$ (Let's call it 50%)

- $R_2 = 0.5 \rightarrow R_2^2 = 0.25$

- **Conclusion:** The first correlation is **twice as good** as the second, because it explains 50% of the variation, while the second only explains 25%. This is *not* obvious from the R-values (0.7 vs 0.5), but *is* crystal clear from the R^2 values (50% vs 25%).

The One Downside of R^2

- R^2 **does not indicate the direction** of the correlation (positive or negative).
- Squaring a negative number (e.g., $R = -0.8$) results in a positive R^2 (e.g., $R^2 = 0.64$).
- **Solution:** You must state the direction if it's not obvious, e.g., "The variables were *negatively correlated* with $R^2 = 0.64$."

VI. Final Summary

1. R^2 is the **percentage of variation** in the Y-variable that is **explained by** the relationship with the X-variable.
2. If someone gives you a value for "plain old R," **square it in your head** to understand the *true* strength and importance of the relationship.